Excretion in Humans

Exam code: 0610 & 0970

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Excretion of carbon dioxide & urea

What is the excretory system?

- The human excretory system is a group of organs which are specialised for the removal of certain excretory products
  - They include the **lungs and kidneys**
  - The **liver** also has a vital role in excretion

Organ	Mainly Excretes	Explanation
Lungs	Carbon dioxide	The lungs excrete carbon dioxide (the waste product of aerobic respiration) during exhalation
Kidneys	Excess water, salts and urea	The kidneys excrete excess water, salts and urea (produced in the liver from excess amino acids) through the formation of urine

What is excretion?

- Excretion is the **removal of the waste substances of metabolic reactions (the chemical reactions that take place inside cells), toxic materials and substances in excess of requirements**
- **Carbon dioxide** must be excreted as it dissolves in water easily to form an acidic solution which can **lower the pH of cells**
  - This can **reduce the activity of enzymes** in the body which are essential for controlling the rate of metabolic reactions
  - For this reason, too much carbon dioxide in the body is **toxic**
- **Urea** is also toxic to the body in higher concentrations and so must be excreted

The excretory system diagram

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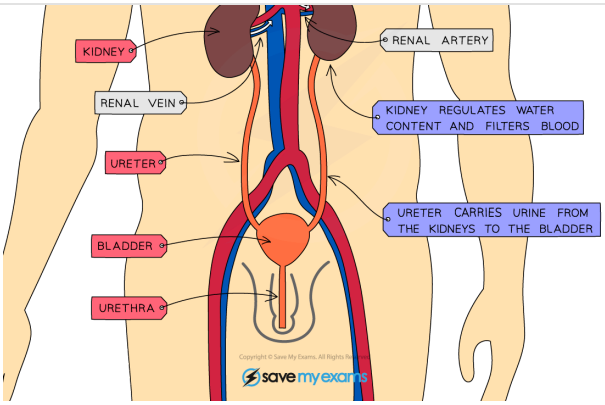

The excretory system in humans

The main components of the excretory system:

Structure	Explanation
Kidney	Two bean-shaped organs that filter the blood
Ureter	Tube connecting the kidney to the bladder
Bladder	Organ that stores urine (excess water, salts and urea) as it is produced by the kidney
Urethra	Tube that connects the bladder to the exterior; where urine is released

Examiner Tips and Tricks

Note the difference between the 'ureter' and the 'urethra'. These two names are commonly confused by students so take care to learn them and know which tube is which – they are NOT interchangeable!

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